KPI Definition Sheet: Hospital Operations & Financial Intelligence

Prepared by Godfrey Miracle | AnalystLab Africa, Data Analytics Track | 3 November 2026

1. How to read this sheet

Each KPI below has a plain-language definition, the DAX measure behind it, the source columns, the value it shows with no filters applied, and the caveat a reader needs to interpret it. The unfiltered values were also calculated independently from the raw CSV files, and they matched the Power BI cards. Every measure recalculates when the Year or Encounter Class slicers change.

2. Core KPIs

KPI	Business question	Definition	DAX measure	Source	Unfiltered value	Caveat
Total Encounters	How much activity did the hospital handle?	Number of encounter records	COUNTROWS(encounters)	encounters	27,891	2022 holds only 220 encounters (1 January to 5 February)
Unique Patients	How many different people were seen?	Distinct patients appearing in the encounters in view	DISTINCTCOUNT(encounters[PATIENT])	encounters.PATIENT	974	Counts patients seen in the current filter, not all patients on file
Total Claim Cost	What is the overall financial size of the activity?	Sum of the total claim cost of every encounter	SUM(encounters[TOTAL_CLAIM_COST])	encounters.TOTAL_CLAIM_COST	101,514,375.52	Includes 89 encounters with a zero claim cost
Avg Claim Cost	What does a typical encounter cost?	Mean total claim cost per encounter	AVERAGE(encounters[TOTAL_CLAIM_COST])	encounters.TOTAL_CLAIM_COST	3,639.68	About 3,651 when the 89 zero-cost encounters are excluded

KPI	Business question	Definition	DAX measure	Source	Unfiltered value	Caveat
Total Payer Coverage	How much did payers pay?	Sum of the amount covered by payers	SUM(encounters[PAYER_COVERAGE])	encounters.PAYER_COVERAGE	31,097,506.99	Zero is recorded both for uninsured patients and for some insured ones
Coverage Rate	What share of cost do payers cover?	Total payer coverage divided by total claim cost	DIVIDE([Total Payer Coverage], [Total Claim Cost])	encounters	30.63%	Cost-weighted, so a few expensive encounters dominate it
Zero Coverage %	How often does the payer cover nothing?	Share of encounters with payer coverage of zero	DIVIDE([Zero Coverage Encounters], [Total Encounters])	encounters.PAYER_COVERAGE	48.71% (13,586 encounters)	8,807 are NO_INSURANCE and 4,779 have an insurer; they mean different things
Over 24h %	How often is an encounter longer than a day?	Share of encounters lasting more than 24 hours	DIVIDE([Over 24h Encounters], [Total Encounters])	encounters.START, STOP	0.27% (75 encounters)	58 of the 75 are inpatient; a few records last months or years
Avg Duration Hours	How long does an encounter last on average?	Mean of STOP minus START, in hours	AVERAGE(encounters[Duration Hours])	encounters.START, STOP	7.27	Distorted by long outliers; the median is 0.25 hours, so read it with

KPI	Business question	Definition	DAX measure	Source	Unfiltered value	Caveat
						Median Duration
Median Duration Hours	How long does a typical encounter last?	Middle value of encounter duration, in hours	MEDIAN(encounters[DurationH ours])	encounters. START, STOP	0.25	Shown on the Encounter Patterns page beside the average
Procedure Count	How many procedures were performed?	Number of procedure records	COUNTR OWS(proc edures)	procedures	47,701	Procedures exist for only 52.6% of encounters
Avg Procedure Base Cost	What does a typical procedure cost?	Mean base cost per procedure record	AVERAG E(procedur es[BASE_ COST])	procedures. BASE_CO ST	2,212.06	Rare, expensive procedures pull the mean up

3. Supporting measures

Measure	Definition	DAX	Unfiltered value
Zero Coverage Encounters	Encounters where payer coverage is zero	SUM(encounters[ZeroCov erage])	13,586
Over 24h Encounters	Encounters longer than 24 hours	SUM(encounters[Over24h])	75
Zero Cost Encounters	Encounters with a total claim cost of zero or less	SUM(encounters[CostIssue Flag])	89
Avg Claim Cost (excl zero)	Average claim cost ignoring the zero-cost records	CALCULATE([Avg Claim Cost], encounters[CostIssueFlag] = 0)	about 3,651
Class Share	One encounter class as a	DIVIDE([Total Encounters],	Ambulatory 44.95%

Measure	Definition	DAX	Unfiltered value
%	share of all encounters	CALCULATE([Total Encounters], ALL(encounters[ENCOUNTERCLASS]))	
Unique Procedure Types	Distinct procedure descriptions	DISTINCTCOUNT(procedures[DESCRIPTION])	163
Encounters with Procedures	Encounters that have at least one procedure	DISTINCTCOUNT(procedures[ENCOUNTER])	14,670
Avg Base Cost (min 20)	Average procedure base cost, shown only for procedures with at least 20 records	IF([Procedure Count] >= 20, [Avg Procedure Base Cost])	Top: Hemodialysis

4. Derived columns used by the measures

Column	Table	Logic
StartDate	encounters	Date part of START, used to link to the date table
DurationHours	encounters	STOP minus START, in hours
Over24h	encounters	1 when DurationHours is above 24, otherwise 0
ZeroCoverage	encounters	1 when PAYER_COVERAGE equals 0, otherwise 0
CostIssueFlag	encounters	1 when TOTAL_CLAIM_COST is 0 or less, otherwise 0
DurationBand	encounters	Over 24 hours, or 24 hours or less
YearQuarter	DateTable	Year and quarter as text, for example 2014-Q1

5. Where each KPI appears

Dashboard page	KPIs shown
Overview	Total Encounters, Unique Patients, Total Claim Cost, Avg Claim Cost, Zero Coverage %, Over 24h %
Encounter Patterns	Over 24h Encounters, Over 24h %, Avg Duration Hours, Median Duration Hours
Cost & Coverage	Total Payer Coverage, Coverage Rate, Zero Coverage %, Zero

Dashboard page	KPIs shown
	Cost Encounters
Procedures	Procedure Count, Unique Procedure Types, Avg Procedure Base Cost, Encounters with Procedures

6. Rules that apply to every KPI

- Records were never deleted. Questionable ones (zero cost, long duration, zero coverage) stay in the totals and are flagged by helper columns.
- 2022 is a partial year and should not be compared with a full year.
- Procedure measures reach the slicers through the link from encounters to procedures, so a Year or Class selection filters them too.
- Percentages are ratios of counts or of sums, never averages of percentages.